Dominick M. Rowan

University Of California, Berkeley

dmrowan@berkeley.edu

0000-0003-2431-981X

POSITIONS HELD

2025 – Present: NASA Hubble Fellow, UC Berkeley

EDUCATION

2020 – 2025: Ohio State University, PhD in Astronomy

Thesis: Characterizing Stars and Compact Objects with Binary Systems in the Era of All-Sky Surveys
Advisor: Krzysztof Z. Stanek

2016 – 2020: Haverford College, Bachelor of Science: Physics & Astronomy

Undergraduate Thesis: *A NICER View of the X-ray Background*
Departmental High Honors
Magna Cum Laude

AWARDS & HONORS

- Allan H. Markowitz Prize (2024): *Recognizes a graduate student who is conducting or developing a research program in areas of observational astronomy, including the analysis of data obtained with space-based observatories, or facilities that are queue-scheduled or provide service observing.*
- Ohio State University Presidential Fellowship (2024 – 2025): *The most prestigious award given by the Graduate School to recognize the outstanding scholarly accomplishments and potential of graduate students entering the final phase of their dissertation research or terminal degree project.*
- David Will Prize in Astronomy (2023): *Recognizes a student or postdoctoral researcher in astrophysics who is conducting an exceptional research program in computational astronomy, survey astronomy, data science, or other related topics.*
- AAS Chambliss Honorable Mention (2023)
- NSF GRFP Honorable Mention (2021)
- Ohio State University Fellowship (2020)
- Louis B. Green Prize in Astronomy (2020)
- Inducted into Phi Beta Kappa Honors Society (2020)
- KINSC Summer Scholar (2017)
- Intel ISEF 1st Place Physics and Astronomy (2016)
- Siemens Competition 5th Place National Finalist (2015)

PUBLICATIONS – 12 as first author, 3 as second author, 9 as contributing author

1. Rowan, D., Kraus, S., Thompson, T. (2025). *Hidden in Plain Sight II: Characterizing the luminous companion to Kappa Velorum with VLTI/GRAVITY.* Submitted to PASP. (`https://ui.adsabs.harvard.edu/abs/2025arXiv251202116R/abstract`)

2. Rowan, D., et al. (2025). *Hidden in Plain Sight: Searching for Dark Companions to Bright Stars with the Large Binocular Telescope and SHARK-VIS*. ApJ. (`https://arxiv.org/abs/2410.19050`)

3. Rowan, D., et al. (2025). *Precise and Accurate Mass and Radius Measurements of Fifteen Galactic Red Giants in Detached Eclipsing Binaries*. The Open Journal of Astrophysics. (`https://arxiv.org/abs/2409.02983`)

4. Rowan, D., et al. (2024). *High mass function ellipsoidal variables in the Gaia Focused Product Release: searching for black hole candidates in the binary zoo*. The Open Journal of Astrophysics. (`https://ui.adsabs.harvard.edu/abs/2024arXiv240109531R/abstract`)

5. Rowan, D., et al. (2024). *A hidden population of massive white dwarfs: two spotted K+WD binaries*. MNRAS. (`https://ui.adsabs.harvard.edu/abs/2024MNRAS.529..587R/abstract`)

6. Rowan, D., et al. (2023). *The Value-Added Catalog of ASAS-SN Eclipsing Binaries III: Masses and Radii of Gaia Spectroscopic Binaries*. MNRAS. (`https://ui.adsabs.harvard.edu/abs/2023MNRAS.523.2641R/abstract`)

7. Rowan, D., et al. (2023). *The Value-Added Catalog of ASAS-SN Eclipsing Binaries II: Properties of Extra-Physics Systems*. MNRAS. (`https://ui.adsabs.harvard.edu/abs/2023MNRAS.520.2386R/abstract`)

8. Rowan, D., et al. (2022). *The Value-Added Catalog of ASAS-SN Eclipsing Binaries: Parameters of Thirty Thousand Detached Systems.* MNRAS (`https://ui.adsabs.harvard.edu/abs/2022MNRAS.517.2190R/abstract`)

9. Rowan, D., et al. (2021). *High Tide: A Systematic Search for Ellipsoidal Variables in ASAS-SN*. MNRAS (`https://ui.adsabs.harvard.edu/abs/2021MNRAS.507..104R/abstract`)

10. Rowan, D., et al. (2020). *A NICER View of Spectral and Profile Evolution for Three X-ray Emitting Millisecond Pulsars*. Astrophysical Journal (`https://ui.adsabs.harvard.edu/abs/2020ApJ...892..150R/abstract`)

11. Rowan, D., et al. (2019). *Detections and Constraints on White Dwarf Variability from Time-Series GALEX Observations*. MNRAS (`https://ui.adsabs.harvard.edu/abs/2019MNRAS.486.4574R/abstract`)

12. Rowan, D., et al. (2016). *The Lick-Carnegie Exoplanet Survey: HD32963b – A New Jupiter-Analog Orbiting a Sun-like Star*. Astrophysical Journal. (`https://ui.adsabs.harvard.edu/abs/2016ApJ...817..104R/abstract`)

Co-Authored Publications – * Indicates a student-led paper

1. *JoHantgen, B., Rowan, D. M., et al. (2025). *A Systematic Search for Big Dippers in ASAS-SN*. Submitted to the Open Journal of Astrophysics. (`https://ui.adsabs.harvard.edu/abs/2025arXiv250719594J/abstract`)

2. *Callahan, J., Rowan, D. M., Kochanek, C. S., Stanek, K. Z. (2025). *Astronomical Cardiology: A Search for Heartbeat Stars Using Gaia and TESS*. Open Journal of Astrophysics. (`https://ui.adsabs.harvard.edu/abs/2025OJAp....8E..97C/abstract`)

3. Forés-Toribio, R., et al. (2025). *ASASSN-24fw: An 8-month long, 4.1 mag, optically achromatic and polarized dimming event.* Submitted to the Open Journal of Astrophysics. (`https://ui.adsabs.harvard.edu/abs/2025arXiv250703080F/abstract`)

4. Kotrach, O., et al. (2024). *Citizen ASAS-SN Data Release II: Variable Star Classification Using Citizen Science*. Submitted to The Open Journal of Astrophysics. (https://ui.adsabs.harvard.edu/abs/2024arXiv240807859K/abstract)

5. Tucker, M. A., Wheeler, A. J., Rowan, D. M., & Huber, M. E. (2024). *Weighing The Options: The Unseen Companion in LAMOST J2354 is Likely a Massive White Dwarf*. The Open Journal of Astrophysics. (https://ui.adsabs.harvard.edu/abs/2024arXiv240719004T/abstract)

6. Phillips, A., et al. (2024). *Seven Classes of Rotational Variables From a Study of 50,000 Spotted Stars with ASAS-SN, Gaia, and APOGEE*. MNRAS. (https://ui.adsabs.harvard.edu/abs/2024MNRAS.527.5588P/abstract)

7. Beck, P. G., et al. (2023). *Red-giant and main-sequence solar-like oscillators in binary systems revealed by ESA Gaia Data Release 3 -- Reconstructing stellar and orbital evolution from binary-star ensemble seismology*. A&A. (https://ui.adsabs.harvard.edu/abs/2024A%26A...682A...7B/abstract)

8. Jayasinghe, T., Rowan, D., et al. (2023). *A search for compact object companions to high mass function single-lined spectroscopic binaries in Gaia DR3.* MNRAS. (https://ui.adsabs.harvard.edu/abs/2023MNRAS.521.5927J/abstract)

9. Olivier, G., et al. (2022). *A Multiwavelength Study of the Massive Colliding Wind Binary WR20a: A possible Progenitor for Fast-Spinning LIGO Binary Black Hole Mergers.* Submitted to ApJ. (https://ui.adsabs.harvard.edu/abs/2022arXiv221202514O/abstract)

10. Jayasinghe, T., et al. (2022). *The 'Giraffe': Discovery of a stripped red giant in an interacting binary with a. 2 $M_\odot$ lower giant*. MNRAS. (https://ui.adsabs.harvard.edu/abs/2022MNRAS.516.5945J/abstract)

11. Jayasinghe, T., et al. (2021) *A unicorn in monoceros: the 3 $M_\odot$ dark companion to the bright, nearby red giant V723 Mon is a non-interacting, mass-gap black hole candidate*. MNRAS. (https://ui.adsabs.harvard.edu/abs/2021MNRAS.504.2577J/abstract)

12. Tucker, M.A., et al. (2018). *ASASSN-18ey: The Rise of a New Black Hole X-ray Binary*. The Astrophysical Journal Letters. (https://ui.adsabs.harvard.edu/abs/2018ApJ...867L...9T/abstract)

White Papers, Research Notes, Classification Reports, and Telegrams:

Rowan, D. M. et al. (2025). ASASSN-25db: A Deep Ongoing Dimming Event. (https://ui.adsabs.harvard.edu/abs/2025ATel17304....1R/abstract)

JoHantgen, B. et al. (2024). ASASSN-25cd: A Deep, Ongoing, Dipping Event. (https://ui.adsabs.harvard.edu/abs/2025ATel17211....1J/abstract)

JoHantgen, B. et al. (2024). ASASSN-25bv: Two Recent, Deep Dimming Events. (https://ui.adsabs.harvard.edu/abs/2025ATel17196....1J/abstract)

JoHantgen, B. et al. (2024). ASASSN-24fw: A Main Sequence Star with a Deep Dimming Event. (https://ui.adsabs.harvard.edu/abs/2024ATel16833....1J/abstract)

JoHantgen, B. et al. (2024). ASASSN-24fa: A Long-Lasting, Deep Dimming Event. (https://ui.adsabs.harvard.edu/abs/2024ATel16715....1J/abstract)

Lam, C. Y., et al. (2023). *Roman CCS White Paper: Characterizing the Galactic population of isolated black holes*. arXiv:2306.12514. (https://ui.adsabs.harvard.edu/abs/2023arXiv230612514L/abstract)

Terry, S. K., et al. (2023). *The Galactic Center with Roman*. arXiv:2306.12485. (https://ui.adsabs.harvard.edu/abs/2023arXiv230612485T/abstract)

Miller, A. M. et al. (2023). ASASSN-23ht: A Deep Eclipse Event. (https://ui.adsabs.harvard.edu/abs/2023ATel16328....1M/abstract)

Rizzo Smith, M. et al. (2022). An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance. (https://ui.adsabs.harvard.edu/abs/2022ATel15531....1R/abstract)

Rizzo Smith, M. et al. (2022). ASASSN-22el: A Deep Eclipse Event. (https://ui.adsabs.harvard.edu/abs/2022ATel15308....1R/abstract)

Rizzo Smith, M. et al. (2021). ASASSN-21sa: A Deep Dimming Event. (https://ui.adsabs.harvard.edu/abs/2021ATel14937....1R/abstract)

Rizzo Smith, M. et al. (2021). ASASSN-21qj: A Rapidly Fading, Sun-Like Star. (https://ui.adsabs.harvard.edu/abs/2021ATel14879....1R/abstract)

Rizzo Smith, M. et al. (2021). ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star. (https://ui.adsabs.harvard.edu/abs/2021ATel14803....1R/abstract)

Rowan, D., et al. (2021). *ASASSN-21co: A Detached Eclipsing Binary with an 11.9 yr Period*. RNAAS (https://iopscience.iop.org/article/10.3847/2515-5172/ac0c83)

Tucker, M.A., Rowan, D.M., & Shappee, B.J. (2018). SCAT Transient Classification Report for 2018-07-22. Transient Name Server Classification Report. (https://ui.adsabs.harvard.edu/abs/2018TNSCR1016....1T/abstract)

Tucker, M.A., Rowan, D.M., & Shappee, B.J. (2018). SCAT Transient Classification Report for 2018-06-13. Transient Name Server Classification Report. (https://ui.adsabs.harvard.edu/abs/2018TNSCR.814....1T/abstract)

Tucker, M.A., Rowan, D.M., & Shappee, B.J. (2018). SCAT Transient Classification Report for 2018-06-12. Transient Name Server Classification Report. (https://ui.adsabs.harvard.edu/abs/2018TNSCR.805....1T/abstract)

Tucker, M.A., et al. (2018). SCAT Classification of 4 Optical Transients. The Astronomer's Telegram. (https://ui.adsabs.harvard.edu/abs/2018ATel11711....1T/abstract)

OUTREACH AND MENTORSHIP

Ohio State Summer Undergraduate Research Program 2024:

Student: Brayden JoHantgen

Project Title: Searching for Big Dipper Stars in ASAS-SN

Current Clemson graduate student

Ohio State Summer Undergraduate Research Program 2023:

Student: Jowen Callahan
Project Title: Determining the Masses and Radii of Heartbeat Stars
Current ASAS-SN Data Analyst

"Polaris" undergraduate physics and astronomy mentorship program, 2021 – 2023

"Polaris is a mentorship program between undergraduate and graduate physics and astronomy students dedicated to fostering a more diverse, equitable, inclusive, and accessible undergraduate physics experience. Our focus lies in the augmented retention of underrepresented and non-traditional groups in the physics and astronomy bachelors of science programs."

"Zenith" SciAccess Blind & Visually Impaired Astronomy Mentorship Program, 2020.

"Program that engages blind and low vision (BLV) students in astronomy and space science by pairing them with student mentors at the Ohio State University. Created in 2020 in partnership with Ohio State and the Ohio State School for the Blind."

Haverford College Public Observing Co-Head, 2019-2020.

Organized observatory open-house events for the local community and private events for small groups/high schools. Led observations with the 12" and 16" telescopes, ran science demonstrations and gave talks.

PRESENTATIONS

Thesis Talk, American Astronomical Society, 245, Jan 2025. National Harbor, MD. *Measuring Precise and Accurate Masses and Radii with ASAS-SN Eclipsing Binaries.*

Talk, Stars & Planets "SPLAT" meeting Sep 2024, University of Hawaii Institute for Astronomy. *Measuring Masses and Radii with ASAS-SN Eclipsing Binaries.*

Contributed Talk, TESS Science Conference III August 2024, Cambridge, MA. *Measuring Fundamental Stellar Parameters with ASAS-SN Eclipsing Binaries.*

Contributed Talk, Cool Stars 22, June 2024, San Diego, CA. *Measuring Fundamental Stellar Parameters with ASAS-SN Eclipsing Binaries*.

Talk, Caltech Tea Talk, April 2024, Pasadena, CA. *Measuring Fundamental Stellar Parameters with Eclipsing Binaries.*

Talk, Penn Journal Club, April 2024, Philadelphia, PA. *Searching for Non-Interacting Black Holes in the Binary Zoo.*

Talk, Princeton Thunch, February 2024, Princeton, NJ. *Searching for Non-Interacting Black Holes in the Binary Zoo.*

Talk, University of Michigan SPF Meeting, November 2023. Ann Arbor, Michigan. *The Value-Added Catalogs of ASAS-SN Eclipsing Binaries.*

Talk, Harvard-Smithsonian Center for Astrophysics ITC Luncheon, March 2023. Cambridge, Massachusetts. *The Value-Added Catalog of ASAS-SN Eclipsing Binaries*

Talk, University of California Santa Barbara, March 2023. *The Value-Added Catalog of ASAS-SN Eclipsing Binaries*

Poster presentation, American Astronomical Society, 241, Jan 2023. Seattle, Washington. *The Value-Added Catalog of ASAS-SN Eclipsing Binaries*

Poster presentation, TESS Science Conference II, Aug 2021. *A Systematic Search for Ellipsoidal Variables with ASAS-SN and TESS*

Poster presentation, American Astronomical Society, 237, Jan 2021. *A Systematic Search for Ellipsoidal Variables in ASAS-SN*

Poster presentation, American Astronomical Society, 235, Jan 2020. Honolulu, Hawaii. *A NICER View of Spectral and Profile Evolution for Three X-ray Emitting Millisecond Pulsars*

Talk, Haverford College Physics & Astronomy Symposium, Dec 2019. *A NICER View of the X-Ray Background*

Poster presentation, Haverford KINSC Summer Research Symposium, September 2019. *Phase-Resolved Spectra of Millisecond Pulsars with NICER X-ray Observations*

Poster presentation, International Pulsar Timing Array, June 2019. Pune, India. *Phase-Resolved Spectra of Millisecond Pulsars with NICER X-ray Observations*

Poster presentation, American Astronomical Society, 234, Jan 2019. *Pulsating and Eclipsing White Dwarfs Discovered from Time-Series GALEX Observations*

Talk, University of Hawaii Institute for Astronomy Summer Research Symposium, August 2018. *White Dwarf Variability in GALEX Observations*.

Talk, Keck Northeast Astronomy Consortium, October 2017. *Mapping Faraday Rotation Measures onto High Velocity Cloud H288*

OBSERVING EXPERIENCE

Large Binocular Telescope: PEPSI, LUCI, LBC, and MODS (18 nights)

Public Observing facilities: Schmidt-Cassegrain 8”, 12”, and 16” telescopes at Haverford

Green Bank 20m radio telescope, 40ft radio telescope

University of Hawaii IfA REU: UH 88inch telescope (4 nights)

ACCEPTED PROPOSALS: 24 as principal investigator, 11 as co-investigator

Transiting Exoplanet Survey Satellite:

“Measuring Fundamental Stellar Parameters of Low-Metallicity Eclipsing Binaries”
PI: Cycle 7 GI – **$70k**

Large Binocular Telescope:
“Observing Non-Interacting Compact Object Companions”
PEPSI, Co-I: 2022A; PI: 2022B, 2023A, 2023B, 2024A, 2024B, 2025A, 2025B– total of 103 hours
“Mass Measurements of Eclipsing Red Giants”
PEPSI, PI: 2023A, 2023B, 2024A – total of 31.2 hours
“Mass Measurements of Low Metallicity Eclipsing Binaries”
PEPSI, PI: 2024B, 2025A, 2025B – 37 hours
“Hidden in Plain Sight: A black hole in the Delta Geminorum system?”
LBTI, PI: 2024B – 2.5 hours
“Hidden in Plain Sight: Direct Imaging of Non-Interacting Black Hole Candidates”
SHARK-VIS, PI: 2024B, 2025A, 2025B – 10 hours
“Measuring dynamical masses of oscillating red giants with SHARK-VIS”
SHARK-VIS, PI: 2025A – 3 hours

Automated Planet Finder Telescope:
“Searching for Non-Interacting Compact Object Companions in Spectroscopic Binaries”
Co-I: 2022B, 2023A, 2023B, 2024B – total of 280 hours
“Determining the masses and radii of evolved stars in detached eclipsing binaries”
Co-I: 2023B – 40 hours
“Finding Black Holes in Gaia’s Spectroscopic Accelerators”
Co-I: 2024A – 70 hours
“Measuring masses and radii of low-metallicity eclipsing binaries”
PI: 2024A, 2024B, 2025A, 2025B – 180 hours
“Uncovering the quiet population of black hole binaries”
Co-I: 2025A, 2025B – 400 hours
“Validating Gaia Orbits of Ellipsoidal Variables”
Co-I: 2025B – 30 hours

Very Large Telescope Interferometer GRAVITY:
“A non-interacting black hole hiding around one of the brightest stars in the sky?”
Co-I: Period 114 – 9 hours
“A non-interacting black hole hiding around a chemically peculiar B-star?”
PI: Period 116 – 3 hours